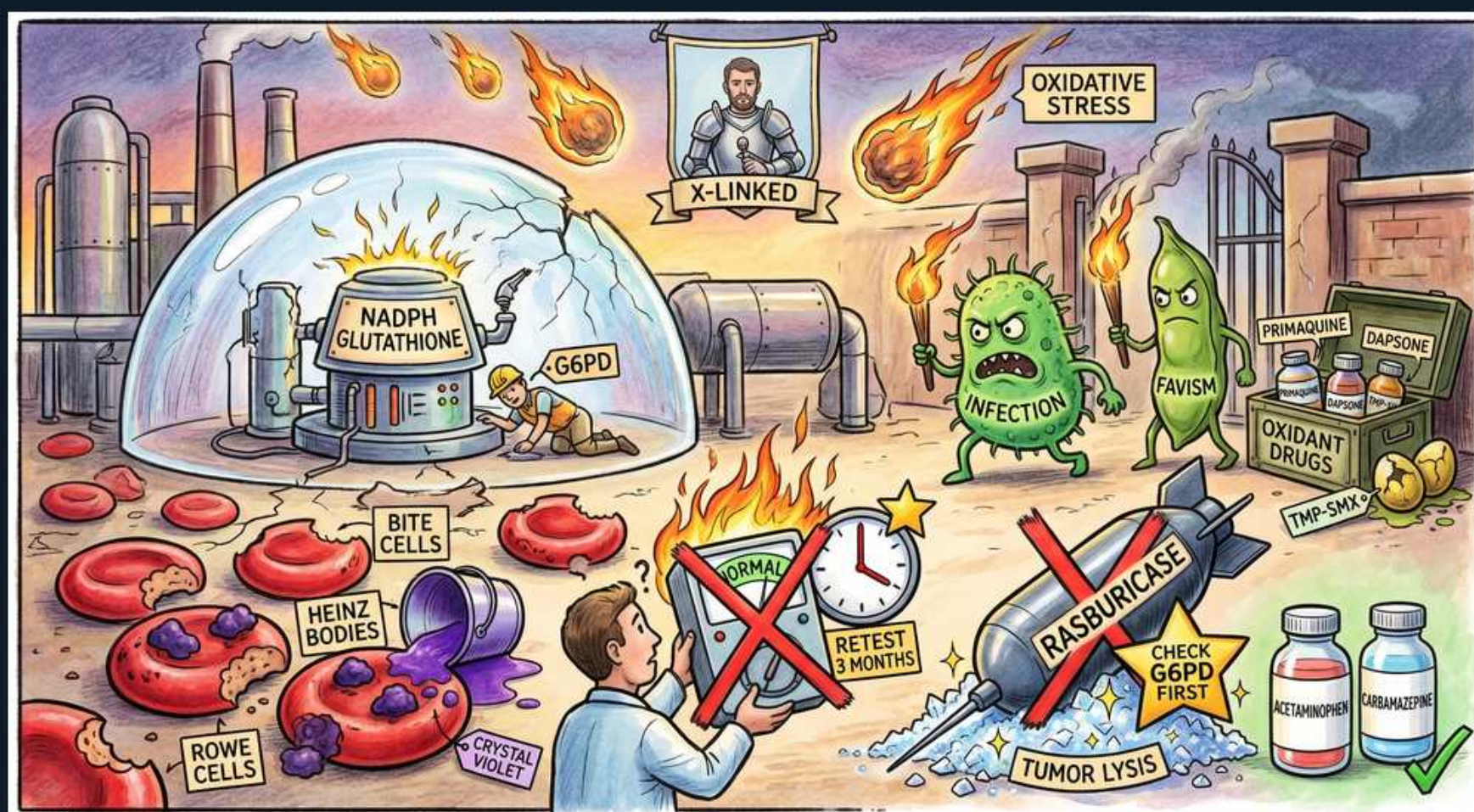


G6PD Deficiency — Smear, Triggers, Drugs & the False-Normal Trap



1. G6PD enzyme

WHAT YOU SEE

The small 'G6PD' valve-tag bolted onto the side of the protective glass dome, the gatekeeper feeding the shield.

WHAT IT ENCODES

G6PD is the rate-limiting enzyme of the hexose-monophosphate (pentose phosphate) shunt, generating NADPH. Its deficiency is the most common human enzymopathy (~400 million people).

BOARD TRAP

Confusing it with pyruvate kinase deficiency — PK is a glycolytic (ATP) defect, whereas G6PD is the HMP shunt (NADPH/antioxidant) defect.

2. NADPH / glutathione shield

WHAT YOU SEE

The big cracked glass dome labeled 'NADPH GLUTATHIONE' arching over the red cells, a shield buckling under the flames raining down on it.

WHAT IT ENCODES

NADPH keeps glutathione reduced, which neutralizes oxidative radicals and protects hemoglobin and membrane. Without it, oxidant stress denatures hemoglobin into Heinz bodies.

3. X-linked

WHAT YOU SEE

The hanging tapestry banner of an armored male knight labeled 'X-LINKED' strung up at the top center.

WHAT IT ENCODES

G6PD deficiency is X-linked recessive, so it predominantly affects males; heterozygous females are usually carriers but can express disease through lyonization.

BOARD TRAP

Calling it autosomal — an affected father cannot pass it to sons. (PK deficiency, by contrast, is autosomal recessive.)

4. Oxidative stress

WHAT YOU SEE

The flaming comet streaking down from the upper-right labeled 'OXIDATIVE STRESS', one of many fireballs assaulting the scene.

WHAT IT ENCODES

Hemolysis in G6PD is EPISODIC, triggered by oxidative stress; between attacks patients are usually well with a normal hemoglobin.

BOARD TRAP

Expecting chronic steady-state anemia — classic G6PD is acute/episodic, not chronic (except rare severe class I variants).

5. Bite cells

WHAT YOU SEE

The cluster of red cells at bottom-left with chunks scooped out of their rims, mislabeled 'ROWE CELLS' (Gemini typo for bite cells / degmacytes).

WHAT IT ENCODES

Splenic macrophages "bite" out the Heinz-body-laden membrane, producing bite cells (degmacytes) and blister cells on the smear.

BOARD TRAP

(with Heinz bodies + a trigger, bite cells point to G6PD)

6. Heinz bodies

WHAT YOU SEE

The dark purple precipitated clumps stuffed inside the red cells, labeled 'HEINZ BODIES' — denatured hemoglobin junk piled in a ketchup-red cell.

WHAT IT ENCODES

Heinz bodies are denatured, precipitated hemoglobin clumped inside the red cell — the fingerprint of oxidative damage.

BOARD TRAP

(see crystal-violet trap)

7. Crystal violet stain

WHAT YOU SEE

The tipped-over bottle of purple dye spilling across the smear, labeled 'CRYSTAL VIOLET' — the special stain needed to reveal the Heinz clumps.

WHAT IT ENCODES

Heinz bodies are invisible on routine Wright-Giemsa; they require a SUPRAVITAL stain (crystal violet or new methylene blue).

BOARD TRAP

"Heinz bodies seen on the routine smear" — they need a supravital stain to be visualized.

8. Infection (trigger)

WHAT YOU SEE

The angry green spiky germ monster labeled 'INFECTION' breathing fire at the red cells, the commonest spark of an attack.

WHAT IT ENCODES

Infection is the MOST COMMON overall trigger of G6PD hemolysis.

BOARD TRAP

Assuming fava beans or drugs are the most common trigger — infection is.

9. Favism

WHAT YOU SEE

The green broad-bean pod character with an angry face labeled 'FAVISM' standing beside the germ, also spitting flames.

WHAT IT ENCODES

Fava (broad) beans cause favism — severe acute hemolysis, especially with the Mediterranean variant, and can be life-threatening in children.

BOARD TRAP

Expecting favism in the mild African A- variant — severe favism is classically the Mediterranean variant.

10. Oxidant drugs

WHAT YOU SEE

The open toolbox/crate of medicine packets labeled 'OXIDANT DRUGS' with tags reading PRIMAQUINE, DAPSONE and TMP-SMX hanging off it.

WHAT IT ENCODES

Oxidant drugs trigger hemolysis: primaquine/tafenoquine, dapsone, sulfonamides (TMP-SMX), nitrofurantoin, methylene blue and rasburicase.

BOARD TRAP

(see rasburicase)

11. Rasburicase — tumor lysis

WHAT YOU SEE

The big missile/bomb labeled 'RASBURICASE' slashed through with a giant red X, a tag reading 'CHECK G6PD FIRST', crashing into the cells below at 'TUMOR LYSIS'.

WHAT IT ENCODES

Rasburicase (recombinant uricase for tumor lysis syndrome) is CONTRAINDICATED in G6PD deficiency — it generates hydrogen peroxide causing severe hemolysis AND methemoglobinemia; screen G6PD before giving.

BOARD TRAP

Giving rasburicase for tumor lysis without G6PD screening — and note methylene blue, the usual methemoglobinemia antidote, is ALSO contraindicated in G6PD.

12. Safe distractors (acetaminophen, carbamazepine)

WHAT YOU SEE

The two harmless-looking pill bottles labeled 'ACETAMINOPHEN' and 'CARBAMAZEPINE' at bottom-right marked with a green check — the safe choices.

WHAT IT ENCODES

These commonly-offered "trigger" distractors are NOT significant oxidant triggers at therapeutic doses — they are safe choices.

BOARD TRAP

Picking acetaminophen / carbamazepine / tacrolimus as the G6PD trigger — the answer is the oxidant drug (rasburicase, primaquine, a sulfonamide).

13. False-normal assay — retest 3 months

WHAT YOU SEE

The gauge/meter dial reading 'NORMAL' crossed out with a big red X, beside a clock labeled 'RETEST 3 MONTHS' — a reading you can't trust mid-attack.

WHAT IT ENCODES

During an acute attack the most-deficient old cells are already lysed and the marrow releases young reticulocytes (which have higher G6PD), so the assay can read FALSELY NORMAL. Retest 2–3 months after recovery, at steady state.

BOARD TRAP

A normal G6PD level during acute hemolysis read as excluding the diagnosis — repeat the assay when the patient is stable.

14. Severity variants

WHAT YOU SEE

The puzzled boy with a '?' over his head standing among the smear, the worried observer weighing how severe this particular variant will be.

WHAT IT ENCODES

Variants differ: Mediterranean (class II, severe), African A- (class III, milder/self-limited because only older cells are affected). Neonatal jaundice on day 2–3 risks kernicterus.

BOARD TRAP

Expecting equal severity — African A- is self-limited/milder; Mediterranean is severe.